Linker Hand L20 Lite
Product Manual



Version revision record

Version	Change date	Change description
V1.0	2025.09.11	Linker Hand L20 Lite First edition

Safety Instructions

Before installing or using this product, please read this manual and related manuals carefully. Be sure to read the relevant descriptions for the safety matters outlined in the manuals.

1. Ensure that this product is connected using the provided or specified cables, and properly secure the product in accordance with regulations. Do not use damaged power cords, plugs, or loose sockets.

2. This product must comply with the installation requirements described in this manual. Using the product beyond the specified installation requirements may shorten the product's service life and could cause serious safety issues.

3. If this product operates at high power for an extended period, its temperature will rise. If the temperature becomes excessively high, stop operation and allow the product to cool down for a period of time before using it again.

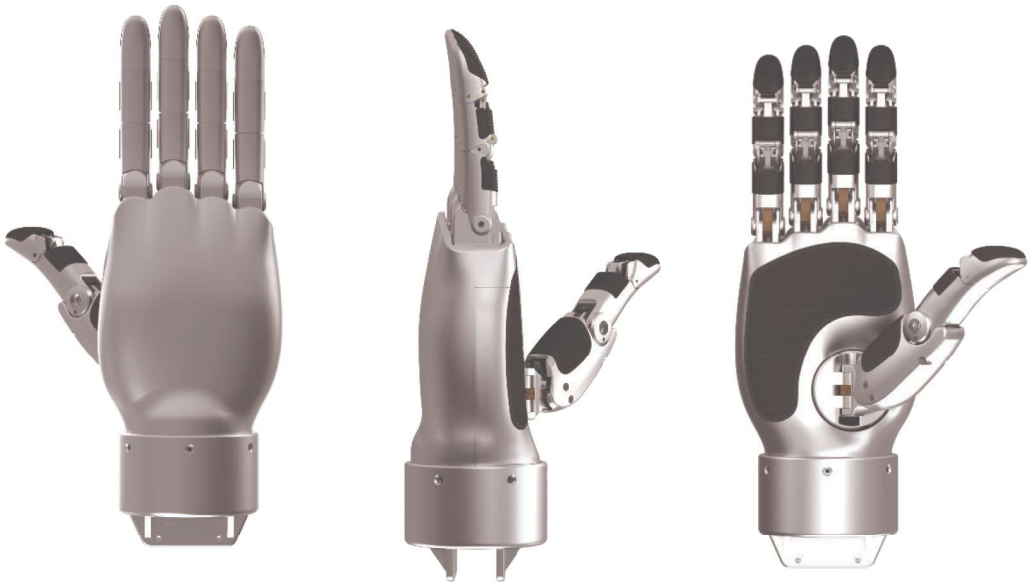
4. This equipment can only be used by trained personnel. Do not open the housing or disassemble the equipment without authorization. If the equipment malfunctions, do not attempt to repair it yourself; please contact our after-sales service department.

5. The diagrams and photos in this manual are representative examples and may differ in details from the purchased product. Additionally, this manual may be revised appropriately due to product improvements, specification changes, or other reasons.

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1 Product Introduction



Product Appearance Diagram

Linker Hand L20 Lite is a high-precision bionic manipulator launched by Linker Bot Company in response to market demands. Driven by self-developed joint motor modules and combined with worm gear transmission, it achieves a high-degree-of-freedom dexterous hand design with 10 Degrees of Freedom (DoF) and 20 joints (10 active joints + 10 passive joints). It features high load capacity, precise control, compact structure and efficient transmission.

1) High Load Capacity

The L20 Lite has excellent load performance: the maximum grasping force of its five fingers can reach 80N, and the maximum fingertip force of both the thumb and the four fingers reaches 15N. This force level enables it to stably grasp and operate objects of various weights, adapting to multi-task scenarios ranging from precision assembly to light industrial operations.

Its high load capacity benefits from the advanced worm gear transmission design. This structure not only provides strong output torque but also has excellent self-locking performance, ensuring stability and safety under load.

2) Precise Control

Precise control is reflected in its high repeat positioning accuracy (less than ±0.2mm) and advanced control system. It supports CAN and RS485 industrial bus interfaces with a communication rate of up to 800kHz, ensuring high-speed, stable transmission of commands and real-time response. Combined with its force-position hybrid control algorithm, the dexterous hand can simulate the fine operations of a human hand, such as precise grasping and flexible pinching, making it suitable for scientific research and industrial scenarios requiring high-precision operations.

3) Compact Structure & Efficient Transmission

The L20 Lite adopts worm gear transmission, featuring a compact structure, large transmission ratio, good self-locking performance (preventing reversal under specific conditions) and stable operation. This allows the dexterous hand to control its size and weight to the greatest extent (total weight of 800g) while ensuring output torque and accuracy. Its configuration of 10 active joints plus 10 passive joints enables complex movements and adaptive grasping.

1.1 Product Functions

1.1.1 Position Control

It can precisely control the spatial position of the entire hand and each finger joint, and smoothly execute preset trajectory movements. This ensures operational accuracy from overall movements to single-joint fine adjustments, meeting scenarios with high position accuracy requirements such as precision assembly and path-based operations.

1.1.2 Speed Control

The finger movement speed can be flexibly adjusted according to task requirements to accurately match different operation rhythms—high-speed mode improves efficiency, while low-speed mode ensures the safety of fine operations. This effectively balances operation efficiency and safety, making it suitable for diverse task scenarios.

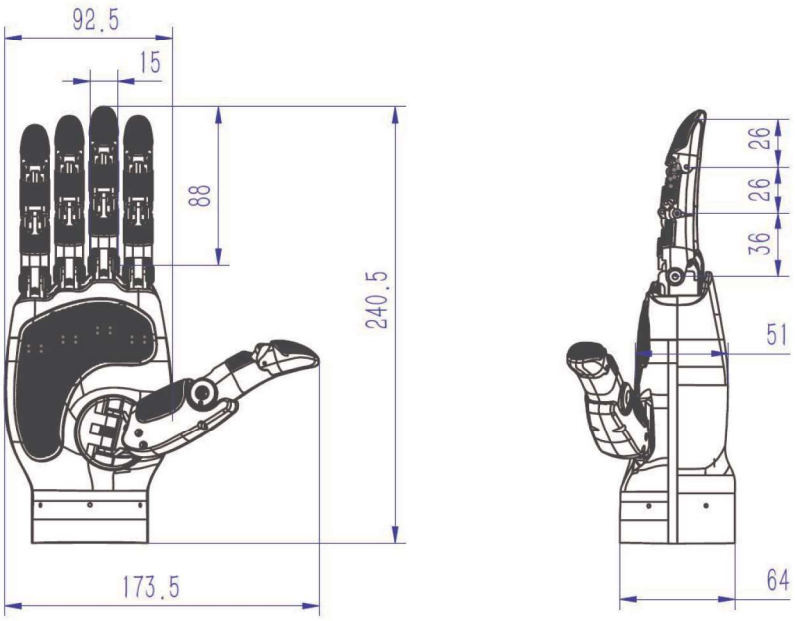
1.1.3 Haptic Feedback (Force Control)

With the help of fingertip sensors, it can real-time detect and precisely control the force and torque applied by the fingers, and dynamically adjust the output force: it not only prevents damage to fragile objects due to excessive force but also avoids object slippage caused by insufficient force, providing reliable force control support for fine operations such as screw tightening and thin sheet grasping.

1.1.4 Online Upgrade

It supports online updates of the dexterous hand system firmware via the host computer, enabling continuous iteration of functional modules and optimization of core performance parameters. It is adaptable to industrial scenarios and scientific research needs that require long-term upgrades and iterations, ensuring the equipment operates efficiently for an extended period.

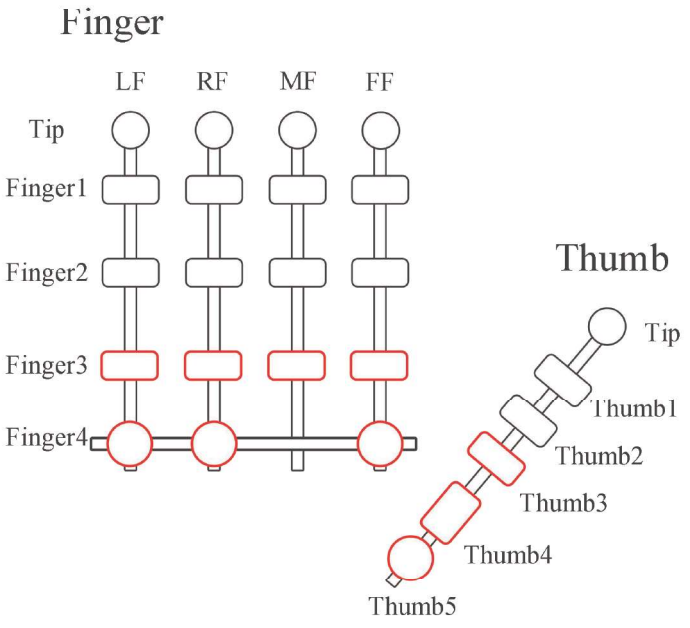
1.2 Appearance Dimensions



Appearance Dimension Drawing (Unit: mm)

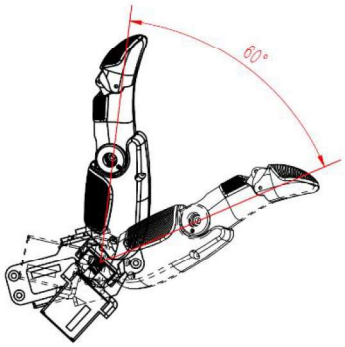
1.3 Degrees of Freedom & Motion Range

This product has a total of 10 Degrees of Freedom (DoF) and 20 joints, including 10 active joints and 10 passive joints. See the following figure for details.

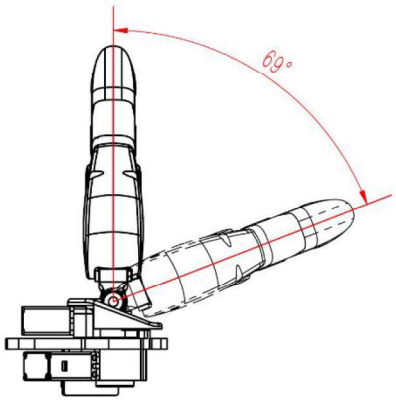


Joint Degrees of Freedom Drawing

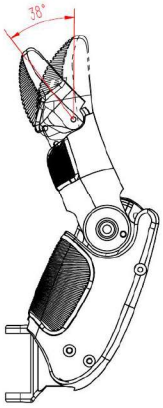
The following is the maximum structural movement range of the product. The actual controlled movement range will be limited to prevent structural collision, so the actual movement range may be smaller than the following range.



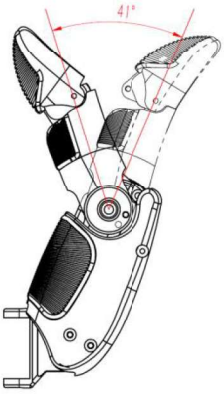
Schematic Diagram of Thumb Rotation Angle



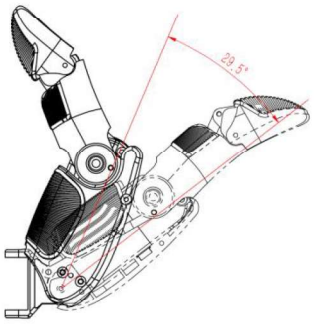
Schematic Diagram of Thumb Side - Swing Angle



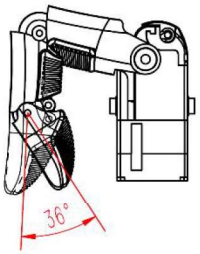
Flexion Angle of the First Thumb Joint



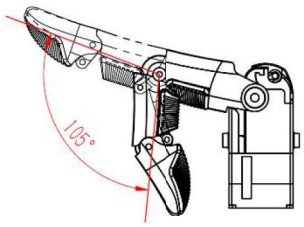
Flexion Angle of the Second Thumb Joint



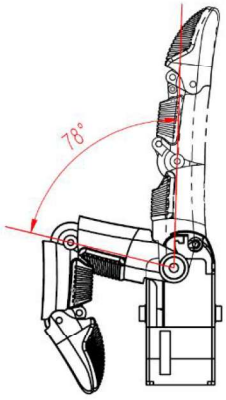
Flexion Angle of the Third Thumb Joint



Flexion Angle of the First Joint of the Four Fingers



Flexion Angle of the Second Joint of the Four Fingers



Full Angle of the Third Joint of the Four Fingers

The following table shows the actual controlled motion angles and motion speeds of this product.

Joints	Maximum Motion Angle (°)	Maximum Motion Speed (°/s)
THUMB1（Flexion）	37.9	63.11
THUMB2（Flexion）	38.4	52.62
THUMB3（Flexion）	27.9	42.78
THUMB4（Lateral Swing）	69	63.21
THUMB5（Rotation）	60	50.9
LF1、RF1、MF1、FF1（Flexion）	35.8	29.11
LF2、RF2、MF2、FF2（Flexion）	102	82.93
LF3、RF3、MF3、FF3（Flexion）	77.9	63.33
FF4（Lateral Swing Toward Thumb）	10.7	84.62
RF4 Side-swing Angle (towards the LF Side)	11.7	44.17
LF4 (towards the lateral side)	17.2	64.93

1.5 Product Parameters

1.5.1 Basic Parameters

Model	Linker hand L20 Lite
Degrees of Freedom (DoF)	10
Number of Joints	20 (10 Active + 10 Passive)
Transmission Method	Linkage Transmission
Control Interface	CAN/RS485
Weight	800g
Maximum Load	25kg
Operating Voltage	DC24V±10%
Static Current	0.2A
Average No-Load Operating Current	0.5A
Maximum Current	3A
Repeat Positioning Accuracy	< ±0.2mm
Opening/Closing Time	1.2s

1.5.2 Force Performance Parameters

Performance Indicator	Specific Parameter
Maximum Thumb Fingertip Force	15N
Maximum Four-Finger Fingertip Force	15N
Maximum Five-Finger Grasping Force	80N

1.6 Sensor System

1.6.1 Tactile Sensor

Linker Hand L20 Lite is equipped with fingertip sensors, which can detect and sense the presence and distance of objects; when in contact, it can accurately capture three-dimensional force and identify surface texture and temperature changes.

1) Pressure Sensor

Parameter	Specification
Piezoresistive Array	6*12
Sensor Force-Bearing Area	9.6*14.4mm
Trigger Force	5g
Measurement Range	20N
Service Life	100,000 Cycles
Communication Frame Rate	200FPS

2) Capacitive Sensor (Optional)

Parameter	Specification
Sampling Frequency	≥50Hz
Measurement Range	0-30N
Overload Limit	60N
Pressure Sensitivity	0.1N
Measurement Resolution	0.5%FS
Measurement Accuracy	2%FS
Pressure Resolution	0.25N
Direction Resolution	45°
Detection Distance	1cm（Metal, Human Body）

1.6.2 Visual-Tactile Perception (Optional)

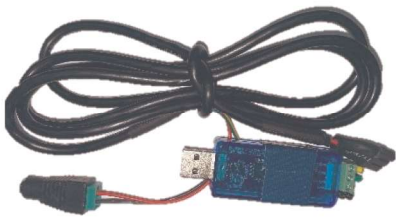
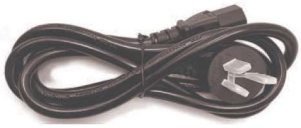
The visual-tactile perception mode essentially integrates visual perception and deep learning large model technology. The principle of this solution is: a high-precision camera captures the deformation of variable flexible materials; when the material is stressed, its surface grid changes shape, and a micro

binocular camera records this deformation process in real-time; then, with the help of a trained deep learning large model, it accurately maps the depth information and movement trend of the object.

2 Installation and Debugging

2.1 Accessories List

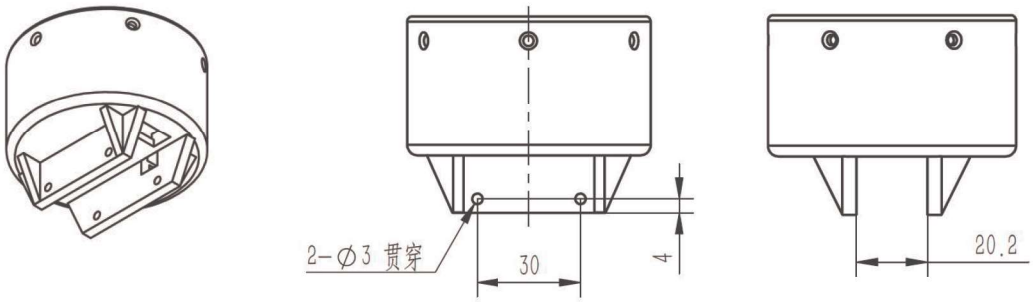
Before installing the dexterous hand, check whether the accessories are complete.

No.	Name	Quantity	Picture
1	USB-to-CAN Debugging Cable	1	
2	Connector Cable XT30 (2+2)	1	
3	Power Adapter	1	
4	Power Cable	1	

2.2 Installation Instructions

2.2.1 Mechanical Installation Instructions

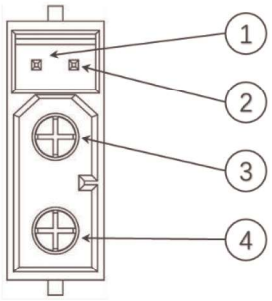
M3 screws are required for installation; the specific length should be adjusted according to the thickness of the fixing part. The installation flange structure diagram is shown below.



Flange Structure Diagram

2.2.2 Electrical Installation Instructions

The dexterous hand uses an XT30 (2+2) connector, and its wire sequence is shown in the following table.



Wire Sequence	Description
1	CAN L
2	CAN H
3	GND
4	24V

2.3 Software Debugging Introduction

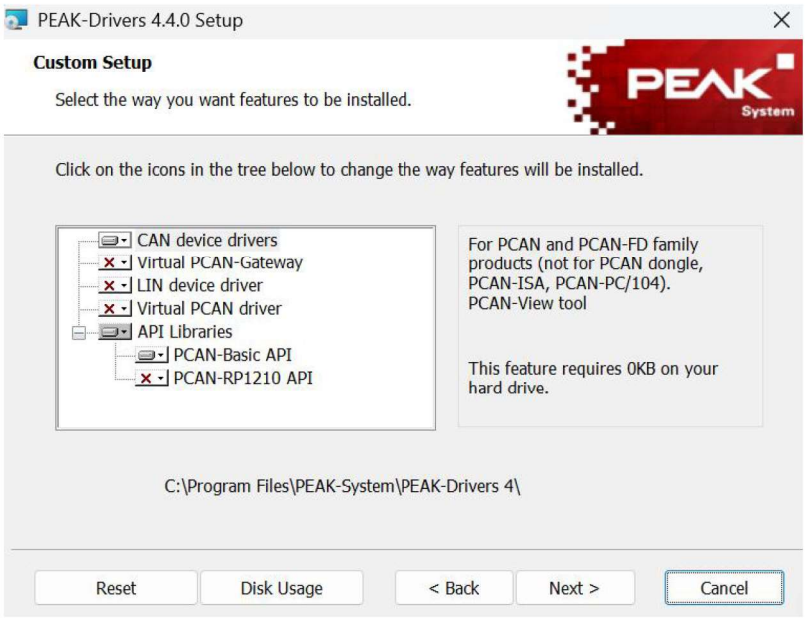
2.3.1 Debugging Preparation

1) Debugging Cable Connection

Use a USB-to-CAN debugging cable to connect the dexterous hand to the debugging PC. Connect the power cable to the power adapter, then plug it into the power port of the USB-to-CAN debugging cable, and plug the other end into an AC220V power strip;

2) USB-to-CAN Driver Installation

Open the file named PEAK-System_Driver-Setup，Install the driver file



step by step (drivers need to be installed for Windows systems, while no installation is required for Linux systems). After the driver program is installed for the first use, there is no need to reinstall it for subsequent use;

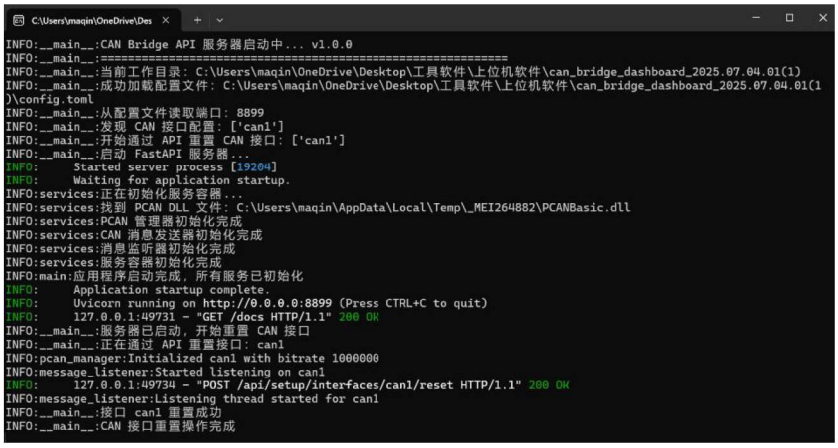
3) Host Computer Software Configuration

Before launching the program, due to the limitations of the Windows system, we need to update the number of connected CAN devices to the configuration file

Open the can_bridge_dashboard folder, modify the can_interface field in the config.toml configuration file, and enter the number corresponding to the number of CAN devices. For example: can_interface = ["can1"] # one device# or can_interface = ["can1", "can2"] # two devices. Save after modification.

4) Start the Adapter Service Program (CAN Bridge)

Double-click to run the can_bridge.exe file：If it runs normally, we will



be able to see the address information of a running service.

If a runtime error occurs, the program will record the exception in the can_bridge.log log file. You can feedback the log content to the after-sales engineer to obtain support information.

Due to the limitations of the Windows system, every time we plug in or unplug a CAN device, we need to close the previous can-bridge program and restart the program.

5) Start the console programs for different dexterous hand devices （Dashboard）

Double-click to run the dashboard.exe file corresponding to the model,



and after waiting for the program to finish running， the console program startup is then completed.

Open http://localhost:9090 in a browser, and you will then be able to access the console normally and control the dexterous hand

2.3.2 Host Computer Software

The main functions of the Linker Hand L20 Lite Host Computer Software include Hand Configuration Management, Global Control, Joint Control, Preset



Actions, Status Monitoring, etc. The page is shown in the figure below.

1) Hand Configuration Management

As shown in the figure below, you can select the port number, hand type, and model, and the online/offline status of the dexterous hand will be



displayed.

2) Global Control

The Global Control functions include Send All Joint Poses, Send All Palm Poses, Send All Joint Speeds, Reset All Hand Components, Stop All Animations, and other functions.

3) Joint Control

In the Joint Control panel, you can perform joint position control, palm control, global speed control, and global torque control.

4) Preset Actions

In the Preset Actions panel, you can preset actions such as making a fist, opening the hand, pinching, and digital gestures. You can also set animations and control the animations.

5) Status Monitoring

In the Status Monitoring panel, you can view the device's real-time logs, hand status, and sensor data, etc.

3 After-Sales Service Terms

1) The product is covered by a 12-month limited warranty from the date of purchase.

2) Exclusion Clauses: The following situations are not covered by the warranty:

- Cosmetic wear and tear caused by normal use.
- Damage caused by improper operation, accidents, unauthorized disassembly, or failure to follow the guidelines.

3) No return or exchange will be accepted for non-quality issues.

4) If the product malfunctions in any way, please contact the official after-sales service immediately and do not disassemble it by yourself.

Contact Us

Contact Type	Details
Website	www.robx.cn
Business Email	Jarvis@robx.cn
Technical Support	support@robx.cn

灵心巧手
LINKERBOT

Linker Hand L20 Lite

产品手册



灵心巧手（北京）科技股份有限公司

版本修订记录

版本号	变更日期	变更说明
V1.0	2025.09.11	Linker Hand L20 Lite 初版

安全须知

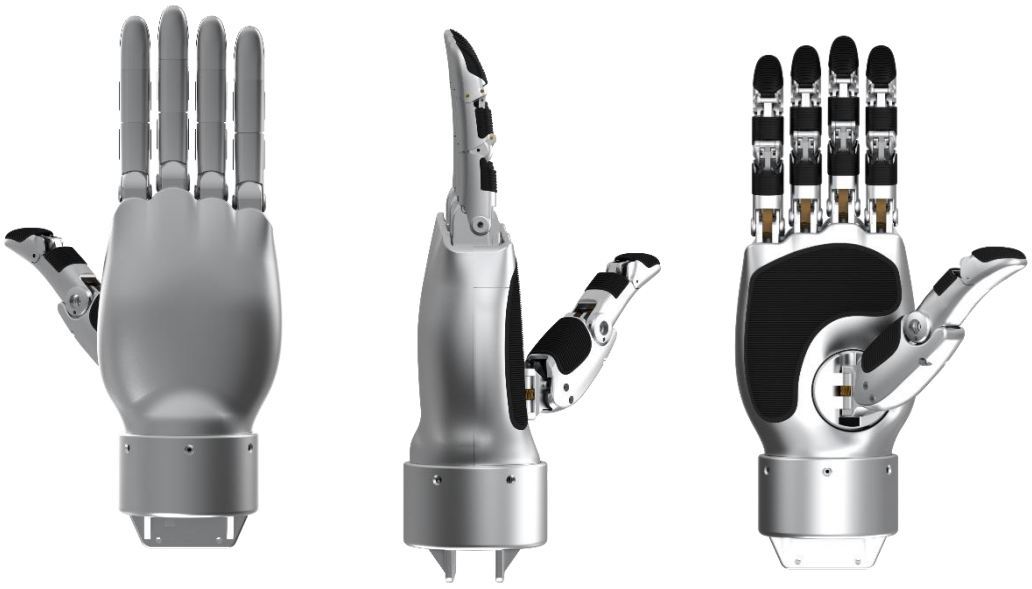
在安装或使用本产品前，请仔细阅读本手册和相关手册，手册中描述得相关安全事项，请务必仔细阅读相关说明。

1. 需确保本产品使用配套或规定的电缆进行连接，并按规定正确固定本产品，勿使用损坏的电源线、插头或松动的插座。
2. 本产品必须符合本手册描述的安装要求，在超出所规定的安装要求条件下使用本产品可能导致产品寿命缩短，并可能造成严重安全问题。
3. 本产品如长时间大功率运行，会出现温度升高的情况，当温度过高，请停止运行，静置冷却一段时间后进行使用。
4. 该设备只能由经过培训的工作人员使用，请勿私自打开外壳以及拆卸。如果设备出现故障，勿自行维修，请与我司售后服务部门联系。
5. 本手册中的图及照片为代表性示例，与所购买产品或有细节差异；另本手册会根据产品改进、规格变更等原因进行适当的修改。

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1 产品介绍



产品外观图

Linker Hand L20 Lite 是灵心巧手公司根据市场需求推出的一款高精度仿生机械手，其采用自研关节电机模组驱动，结合蜗轮蜗杆的传动方式，实现具有 10 个自由度，20 个关节（10 主动关节+10 被动关节）的高自由度灵巧手设计，具有高负载能力、控制精准、结构紧凑传动高效等特点。

1) 高负载能力

L20 Lite 具备出色的负载性能，其五指最大抓握力可达 80N，拇指与四指的最大指尖力均达到 15N。这一力量水平使其能够稳定抓取和操作多种重量的物体，适应从精密装配到轻型工业操作的多任务场景。

其高负载能力得益于先进的蜗轮蜗杆传动设计，这种结构不仅提供了强大的输出力矩，还具有良好的自锁特性，确保了负载下的稳定性和安全性。

2) 控制精准

控制精准体现在其高重复定位精度（小于±0.2mm）和先进的控制系统上。

它支持 CAN 和 RS485 工业总线接口，通信速率高达 800Hz，确保了指令的高速、稳定传输与实时响应。结合其力位混合控制算法，灵巧手能模拟人手的精细操作，如精准抓取、柔性捏握等，适用于需要高精度操作的科研与工业场景。

3) 结构紧凑传动高效

L20 Lite 采用蜗轮蜗杆传动方式，具有结构紧凑、传动比大、自锁性好（在特定条件下可防止逆转）和运行平稳的特点。这使得灵巧手在保证输出力矩和精度的同时，最大程度地控制了体积和重量（整机 800g）。其 10 个主动关节加 10 个被动关节的配置，可实现复杂动作和适应性抓取。

1.1 产品功能

1.1.1 位置控制

能够精准控制手部整体及各手指关节的空间位置，可平稳执行预设轨迹运动，确保从整体动作到单关节微调的操作准确性，满足精密装配、路径化作业等对位置精度要求较高的场景。

1.1.2 速度控制

可根据任务需求灵活调节手指运动速度，精准适配不同作业节奏—高速模式提升效率，低速模式保障精细操作安全，从而高效兼顾作业效率与操作安全性，适配多样化任务场景。

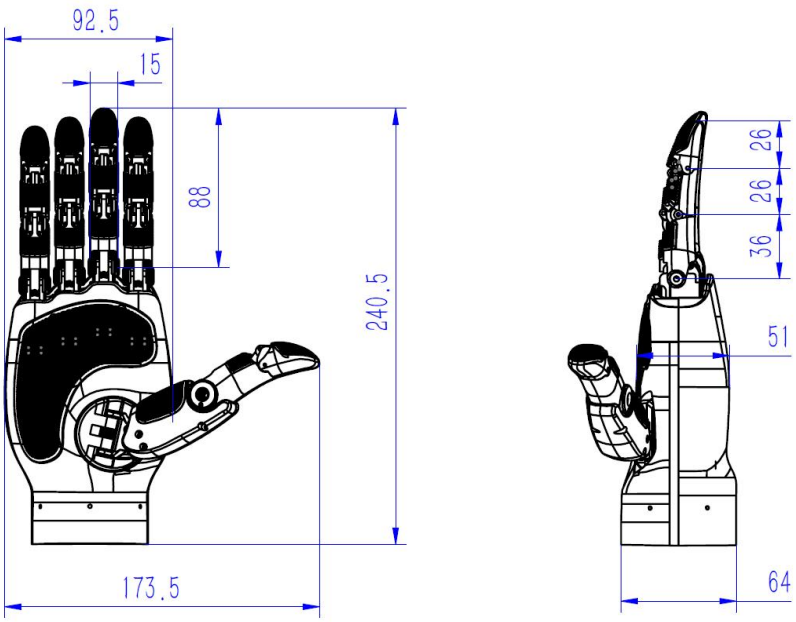
1.1.3 触觉反馈（力控）

借助指尖传感器实时感知并精准控制手指施加的力与扭矩大小，可动态调整输出力度：既避免因力度过大损伤脆弱物体，也防止力度不足导致物体滑落，为螺丝锁附、薄片抓取等精细操作提供可靠力控保障。

1.1.4 在线升级

支持通过上位机对灵巧手系统固件进行在线更新，可持续迭代功能模块、优化核心性能参数，适配需要长期升级迭代的工业场景与科研需求，保障设备长期处于高效运行状态。

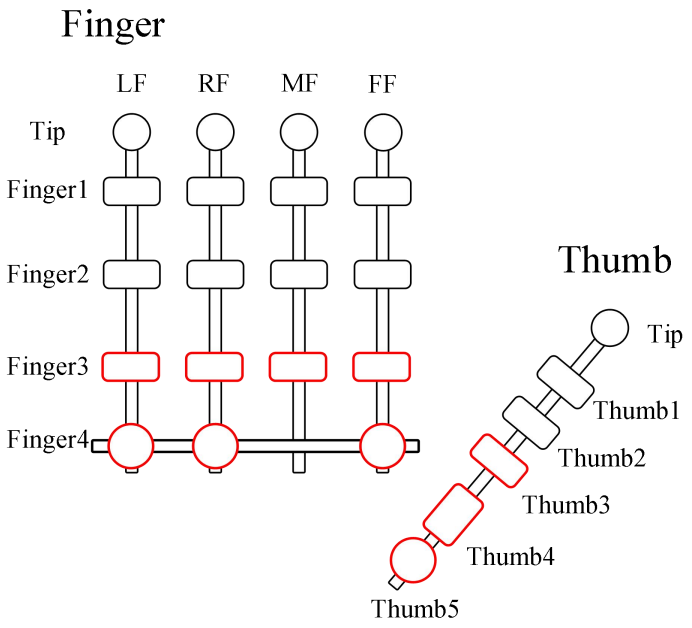
1.2 外观尺寸



外观尺寸图(单位 mm)

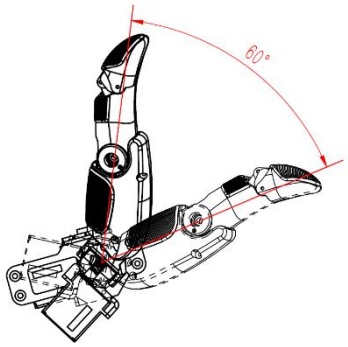
1.3 自由度与运动范围

本产品共 10 个自由度，20 个关节，其中 10 个主动关节，10 个被动关节，具体如下图。

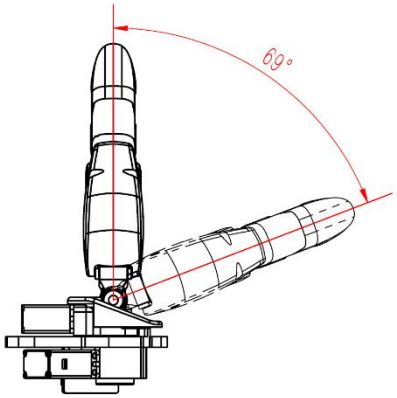


关节自由度图

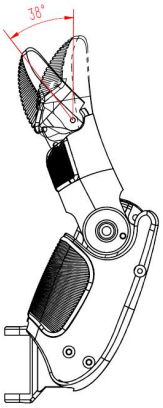
以下为本产品结构活动最大范围，实际控制的活动范围会为了防止结构撞击对活动范围进行限制，因此实际活动范围可能小于以下活动范围。



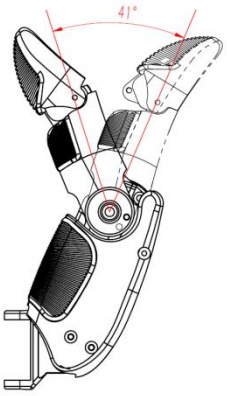
拇指旋转角度示意图



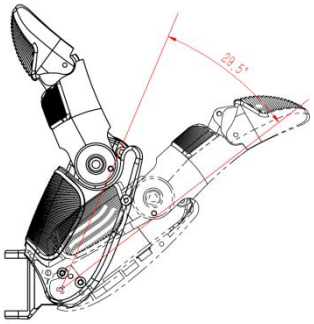
拇指侧摆角度示意图



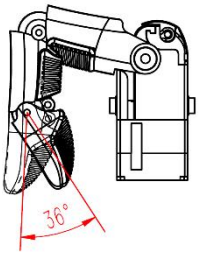
拇指一关节弯曲角度



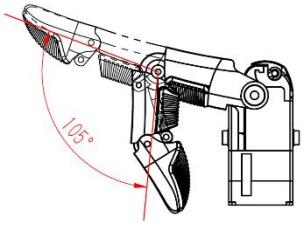
拇指二关节弯曲角度



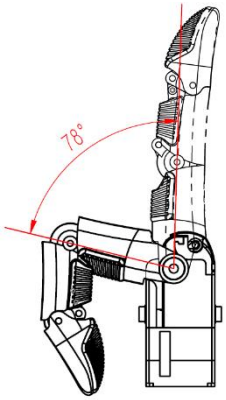
拇指三关节弯曲角度



四指一关节弯曲角度



四指二关节弯曲角度



四指三关节完全角度

下表为本产品实际控制的运动角度和运动速度。

关节(Joints)	最大运动角度(°)	最大运动速度（°/s）
THUMB1（弯曲）	37.9	63.11
THUMB2（弯曲）	38.4	52.62
THUMB3（弯曲）	27.9	42.78
THUMB4（侧摆）	69	63.21
THUMB5（旋转）	60	50.9
LF1、RF1、MF1、FF1（弯曲）	35.8	29.11
LF2、RF2、MF2、FF2（弯曲）	102	82.93
LF3、RF3、MF3、FF3（弯曲）	77.9	63.33
FF4（向 Thumb 侧摆）	10.7	84.62
RF4 侧摆角度（向 LF 侧）	11.7	44.17
LF4（向外侧）	17.2	64.93

1.5 产品参数

1.5.1 基础参数

型号	Linker hand L20 Lite
自由度	10
关节数	20（10 主动 + 10 被动）

传动方式	连杆传动
控制接口	CAN/RS485
重量	800g
最大负载	25kg
工作电压	DC24V±10%
静态电流	0.2A
空载运动平均电流	0.5A
最大电流	3A
重复定位精度	＜±0.2mm
开合时间	1.2s

1.5.2 力性能参数

性能指标	具体参数
拇指最大指尖力	15N
四指最大指尖力	15N
五指最大抓握力	80N

1.6 传感器系统

1.6.1 触觉传感器

Linker Hand L20 Lite 配备指尖传感器，可预测并感知物体的存在与距离；接触时，能精准捕捉三维力，并识别表面纹理与温度变化。

1) 压敏传感器

参数	规格说明
压阻阵列	6*12
传感器受力区域	9.6*14.4mm
触发力	5g
量程	20N
寿命	10 万次
通信帧率	200FPS

2) 电容传感器（选配）

参数	规格说明
采样频率	≥50Hz
测量量程	0-30N
过载极限	60N
压力灵敏度	0.1N
测量分辨率	0.5%FS
测量精度	2%FS
压力分辨率	0.25N
方向分辨率	45°
检测距离	1cm（金属、人体）

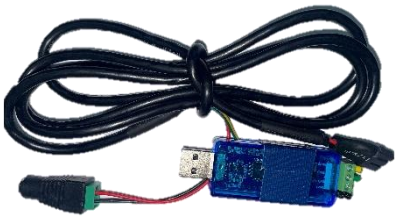
1.6.2 视触觉感知（选配）

视触觉感知模式本质上融合了视觉感知与深度学习大模型技术。该方案的原理是：通过高精度相机捕捉可变柔性材料的形变，当材料受力时，其表面网格会发生形态变化，微型双目相机会实时记录这一形变过程，再借助已训练完成的深度学习大模型，精准映射出物体的深度信息与运动趋势。

2 安装与调试

2.1 配件清单

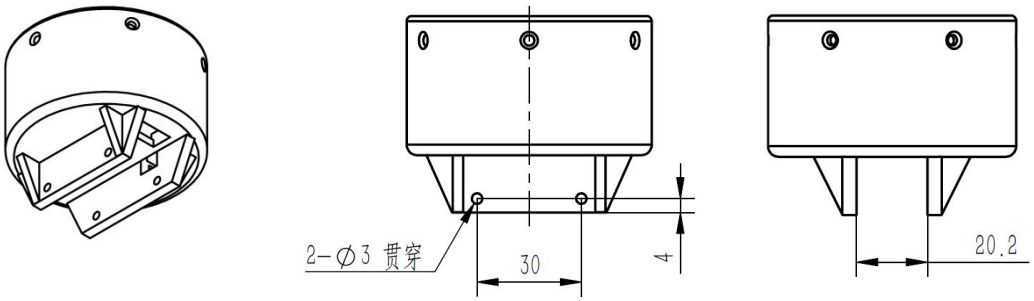
安装灵巧手前，先检查配件是否齐全。

序号	名称	数量	图片
1	USB 转 CAN 调试线	1	
2	电源适配器	1	
3	电源线	1	

2.2 安装说明

2.2.1 结构安装说明

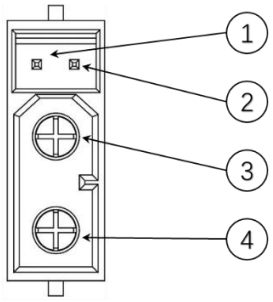
安装需使用 M3 螺丝，具体长度根据固定件厚度调整，安装法兰结构图如下所示。



法兰结构图

2.2.2 电气安装说明

灵巧手采用型号为 XT30（2+2）接插件，其线序如下表。



线序	说明
1	CAN L
2	CAN H
3	GND
4	24V

2.3 软件调试介绍

2.3.1 调试准备工作

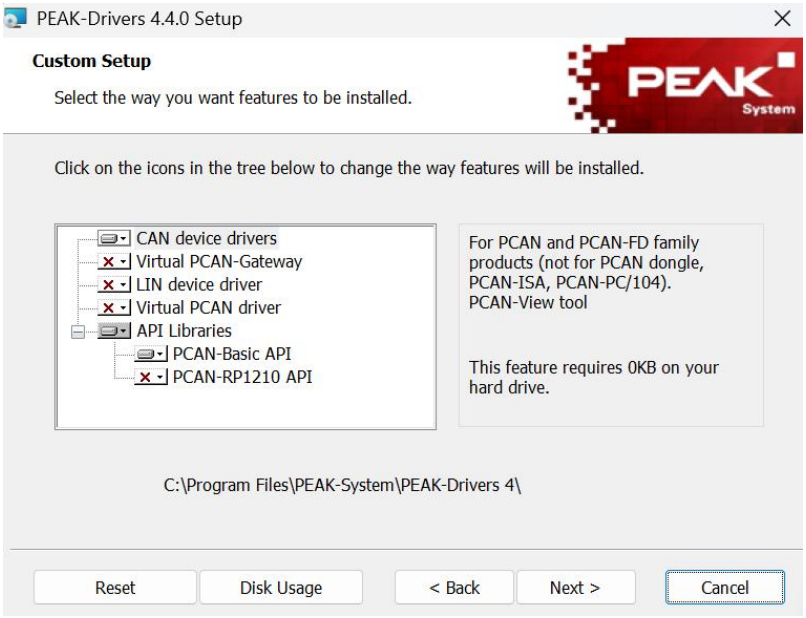
1) 调试线连接

将使用 USB 转 CAN 调试线将灵巧手与调试 PC 连接，将电源线与电源适配器插好后，并插到 USB 转 CAN 调试线的电源口上，另一端插

到 AC220V 插排上；

2) USB 转 CAN 驱动安装

打开文件名 PEAK-System_Driver-Setup 文件，逐步安装驱动文件



（Windows 系统需安装驱动，Linux 系统无需安装），驱动程序初次使用安装后，再次使用无需重复安装；

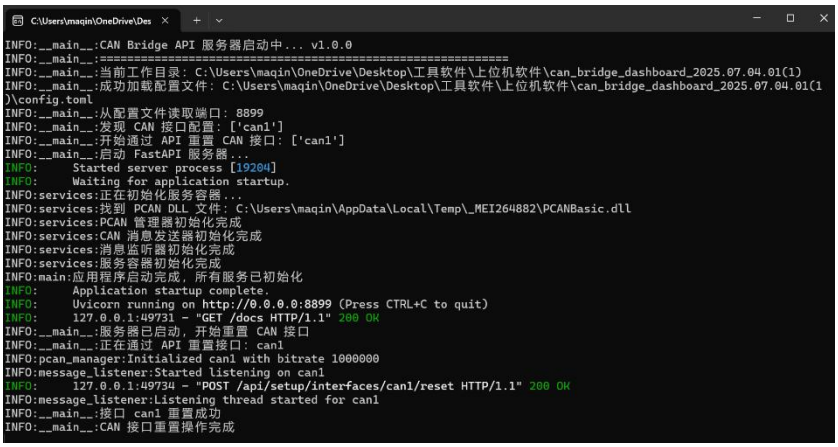
3) 上位机软件配置

在启动程序之前，因为 Windows 系统的限制，我们需要将连接设备的 CAN 数量更新到配置文件中。

打开 can_bridge_dashboard 文件夹，修改 config.toml 配置文件中的 can_interface 字段，有几个 can 设备就写到数字几，比如：can_interface = ["can1"] # 一个设备# 或 can_interface = ["can1", "can2"] # 两个设备，修改后保存。

4) 启动适配器服务程序（CAN Bridge）

双击运行 can_bridge.exe 文件：如果运行正常，我们将能够看到一



个服务运行的地址信息。

如果遇到运行错误，程序将会将异常记录在 can_bridge.log 日志文件中，你可以将日志内容反馈给售后工程师，以获取支持信息。

受限于 Windows 系统，我们每次插拔 can 设备，都需要关闭之前的 can-bridge 程序，并进行程序的重新启动。

5) 启动不同灵巧手设备的控制台程序（Dashboard）

双击运行型号所对应的 dashboard.exe 文件，等待程序运行完毕后，



即完成控制台程序启动。

在浏览器中打开 http://localhost:9090 ，即能够正常访问控制台并

对灵巧手进行控制。

2.3.2 上位机软件说明

Linker Hand L20 Lite 上位机软件主要功能包括手部配置管理、全局控制、关节控制、预设动作、状态监控等，页面如下图所示。



1) 手部配置管理

如下图所示，可选择端口号、手型以及型号，并显示灵巧手在线/离线状态。



2) 全局控制

全局控制功能包括发送所有关节姿态、发送所有掌部姿态、发送所有关节速度、重置所有手部、停止所有动画等功能。

3) 关节控制

在关节控制栏可对关节位置控制、掌部控制、全局速度控制，以及全局扭矩控制。

4) 预设动作

在预设动作栏可预设握拳、张开、捏取以及数字手势等动作，并可设置动画，并对动画进行控制。

5) 状态监控

在状态监控栏可查看设备实时日志、手部状态以及传感器数据等。

3 售后与服务须知

1) 产品享受自购买之日起 12 个月的有限保修。

2) 保修期内，因制造或材料缺陷导致的故障，我们将提供免费维修或部件更换。

3) 免责条款：以下情况不属保修范围：

- 正常使用造成的外观磨损。
- 因不当操作、意外、私自拆卸或未遵循指南导致的损坏。

4) 非质量问题，不予退换。

5) 产品出现任何故障，请立即联系官方售后，切勿自行拆卸。

6) 保修期外的维修将收取相应费用。

附录

ROS_SDK: https://github.com/linker-bot/linkerhand-ros-sdk

ROS2_SDK: https://github.com/linker-bot/linkerhand-ros2-sdk

Python_SDK: https://github.com/linker-bot/linkerhand-python-sdk

技术支持邮箱：support@linkerbot.cn

如需更多产品资料，可访问公司官网：https://linkerbot.cn

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